Flip-Flop Conversions

The Golden Methodology (4 Steps)

To convert any flip-flop into another type, you are essentially designing a combinational logic circuit that sits in front of the **Available** flip-flop to make it behave like the **Desired** flip-flop.

You must always follow these 4 steps:

1. **Identify:** State what you *have* (Available FF) and what you *want* (Desired FF).
2. **The Conversion Table:** Create a master table. The inputs are the pins of the Desired FF and the current state (Q_n). Use the Desired FF's Characteristic logic to find the Next State (Q_{n+1}). Then, use the Available FF's Excitation Table to figure out what physical inputs are required to make that Q_{n+1} happen.
3. **K-Map Derivation:** Plot the required inputs onto K-Maps to find their simplified Boolean equations.
4. **The Circuit Diagram:** Draw the Available FF and wire its inputs using the logic gates derived in Step 3.

Conversion 1: SR to D Flip-Flop

- **Desired:** D Flip-Flop
- **Available:** SR Flip-Flop

Step 2: The Conversion Table

We use the D Flip-Flop logic ($Q_{n+1} = D$) to find the next state. Then we use the SR Excitation Table to figure out what S and R need to be to cause that transition.

Desired Input	Current State	Next State	Required Available Inputs	
D	Q_n	Q_{n+1}	S	R
0	0	0	0	X
0	1	0	0	1
1	0	1	1	0
1	1	1	X	0

Step 3: K-Map Derivation

From your conversion table, we plot the required inputs S and R onto separate 2-variable Karnaugh maps based on the inputs D and Q_n.

K-Map for S:

$D \backslash Q_n$	0	1
0	0	0
1	1	X

- **Grouping:** We group the `1` and the Don't Care (`X`) in the bottom row ($D = 1$).
- **Equation:** $S = D$

K-Map for R:

$D \backslash Q_n$	0	1
0	X	1
1	0	0

- **Grouping:** We group the `X` and the `1` in the top row ($D = 0$).
- **Equation:** $R = \overline{D}$

If we plot S and R onto separate 2-variable K-Maps (for D and Q_n), we get very simple loops:

- **For S:** A loop of 1s in the $D = 1$ row. Equation:

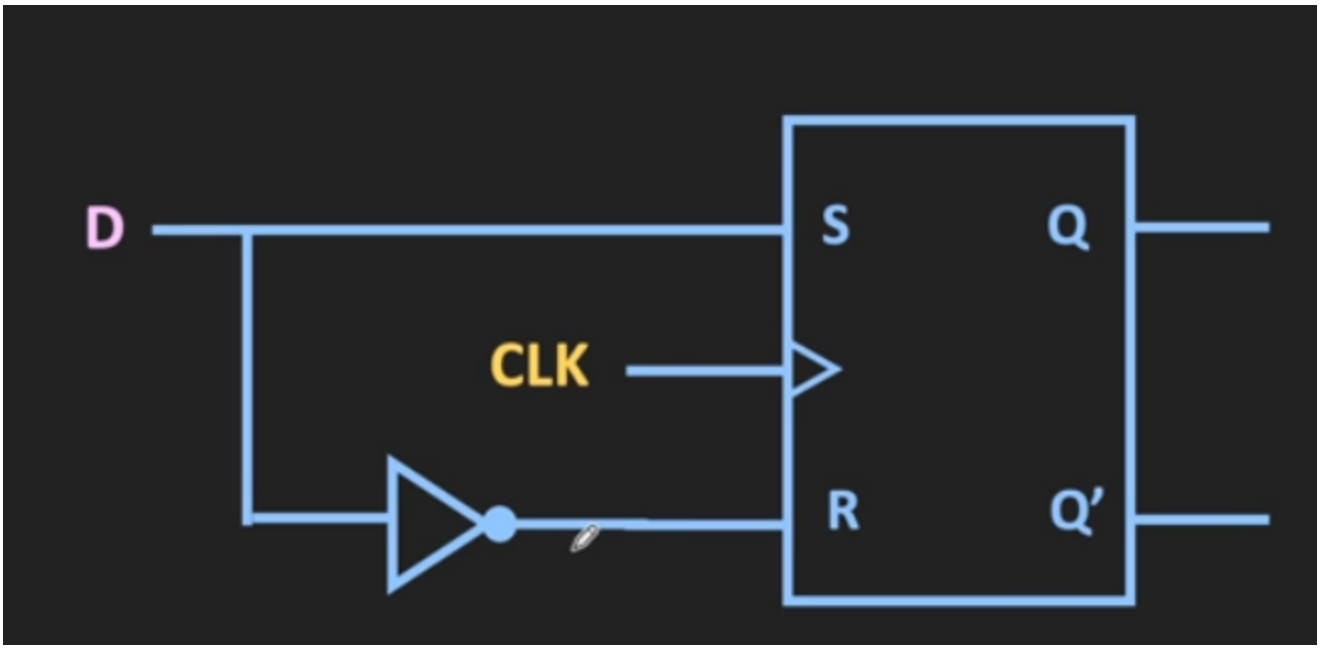
$$S = D$$

- **For R:** A loop of 1s in the $D = 0$ row. Equation:

$$R = \overline{D}$$

Step 4: Circuit Diagram

This mathematically proves the D-Latch/Flip-Flop architecture we covered previously. We simply attach an inverter to the D input and route it to R.



Conversion 2: T to JK Flip-Flop

- **Desired:** JK Flip-Flop
- **Available:** T Flip-Flop

Step 2: The Conversion Table

We use the JK Characteristic logic to determine Q_{n+1}. Then we look at the T Excitation logic (remember, $T = 1$ if the state changes, $T = 0$ if it holds) to find the required T input.

Desired Inputs		Current	Next State	Req. Input
J	K	Q_n	Q_{n+1}	T
0	0	0	0	0
0	0	1	1	0
0	1	0	0	0
0	1	1	0	1
1	0	0	1	1
1	0	1	1	0
1	1	0	1	1
1	1	1	0	1

Step 3: K-Map Derivation

We plot the required T input onto a 3-variable K-Map based on the desired inputs (J, K) and the current state (Q_n).

K-Map for T:

$J\backslash KQ_n$	00	01	11	10
0	0	0	1	0
1	1	0	1	1

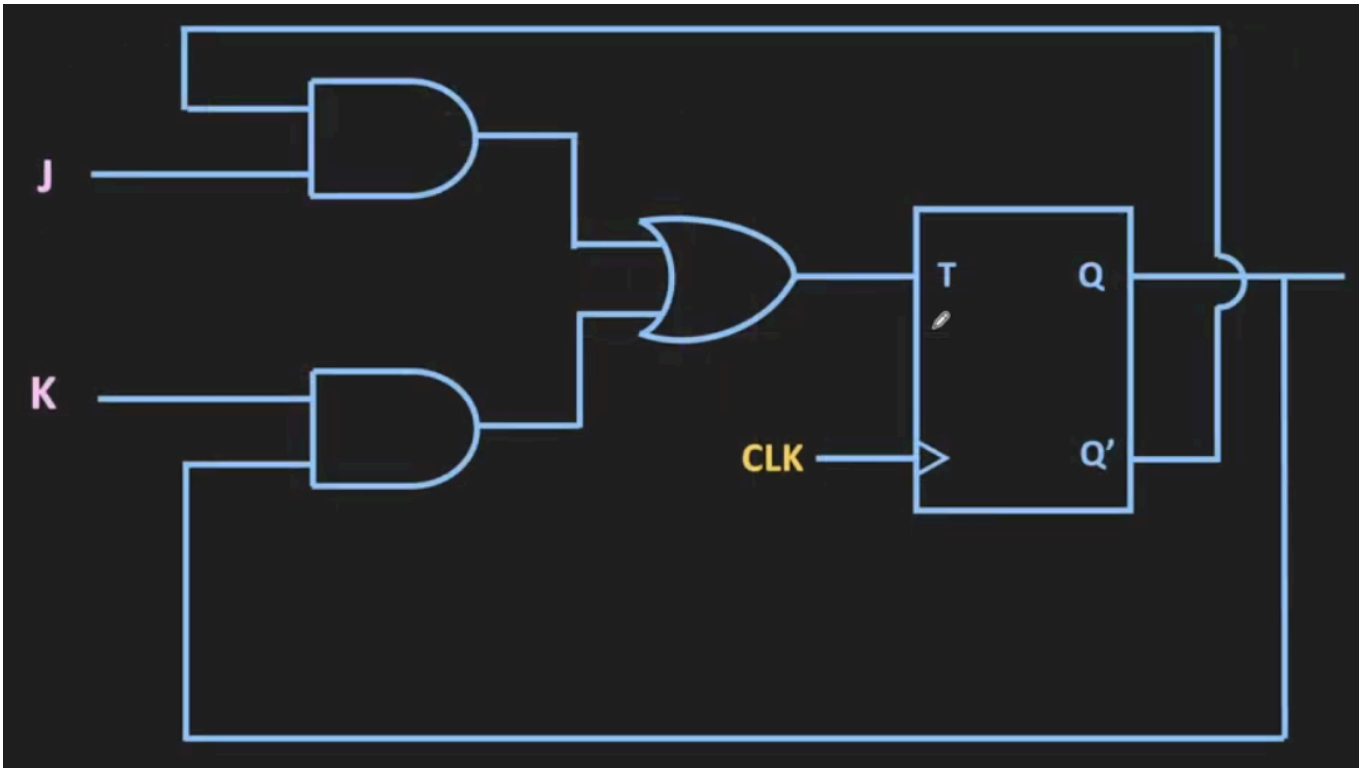
- **Grouping 1:** Group the `1`s at `100` and `110`. This eliminates K. The remaining term is $J\overline{Q}_n$.
- **Grouping 2:** Group the `1`s at `011` and `111`. This eliminates J. The remaining term is KQ_n.
- **Equation:** $T = J\overline{Q}_n + KQ_n$
- Group 1 groups the 1s where $J = 1$ and $Q_n = 0$.
- Group 2 groups the 1s where $K = 1$ and $Q_n = 1$.

Equation:

$$T = J\overline{Q}_n + KQ_n$$

Step 4: Circuit Diagram

We must use two AND gates and an OR gate to feed the correct logic into the T pin.



Conversion 3: SR to JK Flip-Flop

- **Desired:** JK Flip-Flop
- **Available:** SR Flip-Flop

Step 2: The Conversion Table

We map all 8 possible combinations of J, K, Q_n to find Q_{n+1}, and then use the SR Excitation logic to map out the required S and R inputs.

Desired Inputs		Current	Next State	Req. Inputs	
J	K	Q_n	Q_{n+1}	S	R
0	0	0	0	0	X
0	0	1	1	X	0
0	1	0	0	0	X
0	1	1	0	0	1
1	0	0	1	1	0
1	0	1	1	X	0
1	1	0	1	1	0
1	1	1	0	0	1

Step 3: K-Map Derivation

K-Map for S:

$J \backslash KQ_n$	00	01	11	10
0	0	X	0	0
1	1	X	0	1

- **Grouping:** We loop the `1`s at `100` and `110`. We *ignore* the `X`s in the `01` column because looping them wouldn't help us form a valid 4-cell group with the existing `1`s.
- **Equation:** $S = J\overline{Q}_n$

K-Map for R:

$J \backslash KQ_n$	00	01	11	10
0	X	0	1	X
1	0	0	1	0

- **Grouping:** We loop the `1`s at `011` and `111`. We ignore the `X`s in the `00` and `10` columns.
- **Equation:** $R = KQ_n$
- **For S:** We group the 1s and Xs where $J = 1$ and $Q_n = 0$Equation:

$$S = J\overline{Q}_n$$

- **For R:** We group the 1s and Xs where $K = 1$ and $Q_n = 1$Equation:

$$R = KQ_n$$

Step 4: Circuit Diagram

Look closely at this final diagram. By routing the outputs Q and Q' back to the front via AND gates, we have mathematically proven *exactly* how the original JK Flip-Flop was invented to solve the SR Flip-Flop's invalid state!

